Open the evidence document **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: evidence_zz999_9999

One source file is used to answer **Question 1** and two source files are used to answer **Question 3**. The files are called **`Data.txt`**, **`AnimalData.txt`** and **`ColourData.txt`**

A class declaration can be used to declare a record.

A list is an alternative to an array.

1 A program reads data from a file and searches for specific data.

(a) The main program needs to read 25 integer data items from the text file `Data.txt` into a local 1D array, `DataArray`

(i) Write program code to declare the local array `DataArray`

Save your program as **Question1_J2023**. Copy and paste the program code into **part 1(a)(i)** in the evidence document.

[1]

(ii) Amend the main program to read the contents of `Data.txt` into `DataArray`

Save your program. Copy and paste the program code into **part 1(a)(ii)** in the evidence document.

[4]

(b) **(i)** The procedure `PrintArray()` takes an integer array as a parameter and outputs the contents of the array in the order they are stored.

The items are printed on the same line, for example:

10 4 5 13 25

Write program code for the procedure `PrintArray()`

Save your program. Copy and paste the program code into **part 1(b)(i)** in the evidence document.

[3]

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(ii) Amend the main program to output the contents of `DataArray` using the procedure `PrintArray()`

Save your program. Copy and paste the program code into **part 1(b)(ii)** in the evidence document.

[1]

(iii) Test your program.

Take a screenshot of the output.

Save your program. Copy and paste the screenshot into **part 1(b)(iii)** in the evidence document.

[1]

(c) The function `LinearSearch()`:

- takes an integer array and integer search value as parameters
- counts and returns the number of times the search value is found in the array.

Write program code for the function `LinearSearch()`

Save your program. Copy and paste the program code into **part 1(c)** in the evidence document.

[3]

(d) (i) Amend the main program to:

- prompt the user to input a whole number between 0 and 100 inclusive
- read and validate the input from the user
- call `LinearSearch()` with `DataArray` and the validated input value
- output the result in the format:
  `The number 7 is found 2 times.`

Save your program. Copy and paste the program code into **part 1(d)(i)** in the evidence document.

[4]

(ii) Test your program by inputting the number 12.

Take a screenshot of the output.

Save your program. Copy and paste the screenshot into **part 1(d)(ii)** in the evidence document.

[1]